

Education

Ph.D. in Computer Science , University of Waterloo, Canada	Sept. 2021 — Oct. 2025
- Advisors: Dr. Naomi Nishimura and Dr. Amer E. Mouawad - Thesis: “On Token Movement Problems”	
B.Sc. in Computer Science , American University of Beirut, Lebanon	Jan. 2019 — May 2021
Graduated with High Distinction, GPA: 4.0/4.0	
B.B.A. - with a focus on Finance, American University of Beirut, Lebanon	Sept. 2013 — Dec. 2016

Scholarships and Awards

Computer Science PhD Student Award , University of Waterloo [3,000 CAD]	2025 — 2026
International Doctoral Student Award , University of Waterloo [60,000 CAD]	2022 — 2025
Mathematics International Master’s Award of Excellence , University of Waterloo [7,500 CAD]	2021 — 2022
University of Waterloo Graduate Scholarship , University of Waterloo [4,000 CAD]	2021 — 2022
Henri Qais Naccache Computer Science Student Development Fund , American University of Beirut [500 USD]	2020 — 2021
United States AID University Scholar , American University of Beirut [125k+ USD]	2013 — 2016

Conference Publications (*Alphabetical Authorship)

[ISAAC’24]*	Mario Grobler, Stephanie Maaz , Amer E. Mouawad, Naomi Nishimura, Vijayaragunathan Ramamoorthi, and Sebastian Siebertz. Kernelization Complexity of Solution Discovery Problems , The 35th International Symposium on Algorithms and Computation, 2024 [pdf]
[ICALP’24]*	Mario Grobler, Stephanie Maaz , Nicole Megow, Amer E. Mouawad, Vijayaragunathan Ramamoorthi, Daniel Schmand, and Sebastian Siebertz. Solution Discovery via Reconfiguration for Problems in P , The 51st EATCS International Colloquium on Automata, Languages, and Programming, 2024 [pdf]
[WALCOM’22]*	Alexandre Cooper, Stephanie Maaz , Amer E. Mouawad, and Naomi Nishimura. Parameterized Complexity of Reconfiguration of Atoms , The 16th International Conference and Workshops on Algorithms and Computation, 2022 [pdf]

Journal Publications (*Alphabetical Authorship)

[Algorithmica’24]*	Alexandre Cooper, Stephanie Maaz , Amer E. Mouawad, and Naomi Nishimura. Parameterized Complexity of Reconfiguration of Atoms , Algorithmica, 2024 [pdf]
[PRA’23]	Remy El Sabeh, Jessica Bohm, Zhiqian Ding, Stephanie Maaz , Naomi Nishimura, Izzat El Hajj, Amer E. Mouawad, and Alexandre Cooper. Efficient Algorithms to Solve Atom Reconfiguration Problems. II. Assignment-Rerouting-Ordering Algorithm , Physical Review A, 2023 [pdf]
[PRA’23]	Barry Cimir, Remy El Sabeh, Marc Bacvanski, Stephanie Maaz , Izzat El Hajj, Naomi Nishimura, Amer E. Mouawad, and Alexandre Cooper. Efficient Algorithms to Solve Atom Reconfiguration Problems. I. Redistribution-Reconfiguration Algorithm , Physical Review A, 2023 [pdf]

Manuscripts (*Alphabetical Authorship)

2025*	Nicolas Bousquet, Stephanie Maaz , Amer E. Mouawad, Naomi Nishimura, and Sebastian Siebertz. On Algorithmic Meta-Theorems for Solution Discovery: Tractability and Barriers , Preprint, arXiv:2510.17344, 2025 [pdf]
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Algorithms and Complexity

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Academic Service

External Reviewer	Workshop on Graph-Theoretic Concepts in Computer Science [WG'23]
External Reviewer	International Colloquium on Automata, Languages, and Programming [ICALP'25]

Contributed Talks

Nov. 2024	Kernelization Complexity of Solution Discovery Problems, Mohamed bin Zayed University of AI (MBZUAI) , Abu Dhabi, UAE.
Jan. 2024	Parametrized Complexity of Reconfiguration of Atoms, 1st Villeurbanne Reconfiguration Workshop (VRW'24) , Lyon, France.
May 2022	Parametrized Complexity of Reconfiguration of Atoms, Banff International Research Station (BIRS'22) , Banff, Canada.

Service

2023 — 2025	Senator, Graduate Student Representative , University of Waterloo, Canada Gained exposure to university governance, maintained relationships with graduate student bodies, and actively participated in Senate meetings.
2015 — 2017	Community Service Organizer , U.S. Agency for International Development , Lebanon Worked on a team to secure funding and collaborate with the local communities to open an ancient cultural wilderness trail which entailed mapping the trail, clearing its pathways, establishing signage, and documentation through project reports.
2013 — 2017	Community Service Volunteer , CCECS, American University of Beirut , Lebanon Led math tutorials for Palestinian refugees preparing for standardized tests (SAT, ACT), and tutored over 15 Syrian refugees at the NASMA Learning and Resource Center to improve academic and social skills.
2014	Community Service Volunteer , D-RASATI - Education Development Center , Lebanon Participated in a nationwide assessment of Lebanese public schools, interviewed principals and evaluated technological readiness to equip computer and science laboratories in over 300 schools.

Experience

Teaching Assistant , University of Waterloo, Canada <i>Courses</i> : CS245 Logic and Computation, CS251 Computer Organization & Design	Sept. 2021 — Aug. 2024
Financial Analyst , PFC International , Lebanon <i>Focus</i> : Strategic planning & advisory, investment analysis and market research	Oct. 2017 — Dec. 2018
Intern , Business Engineering Experts , Lebanon <i>Focus</i> : Corporate governance, financial model design and cost of equity estimation	May. 2017 — Jun. 2017
Intern , Byblos Bank Group , Lebanon <i>Focus</i> : Compliance with key regulatory objectives such as FATCA and confidentiality	Jun. 2016 — Jul. 2016